

CENG381 Stochastic Processes

Renewal-Reward and Residual Life — Practice 1

Q1 (30 points). The time-average residual life (expected wait for the next renewal when arriving at a random time) is given by:

$$Y = E[X^2] / (2 \cdot E[X])$$

Compute Y for each of the following inter-renewal distributions and compare with $E[X]/2$.

- a) Deterministic: $X = 6$ minutes (every renewal happens exactly 6 minutes apart). Compute Y . Explain geometrically why $Y = E[X]/2$ in this case.
 - b) Exponential: $X \sim \text{Exp}(1/6)$ (rate $1/6$, mean 6 minutes). Recall $E[X^2] = 2/\lambda^2$ for $\text{Exp}(\lambda)$. Compute Y . Why is $Y = E[X]$ rather than $E[X]/2$ for an exponential distribution?
 - c) Uniform: $X \sim \text{Uniform}[0, 12]$ (mean 6 minutes). Recall $E[X^2] = (b^3 - a^3) / (3(b - a))$ for $\text{Uniform}[a, b]$. Compute $E[X^2]$ and Y . Is Y greater or less than $E[X]/2$? Explain the inspection paradox.
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Q2 (35 points). The inspection paradox: the renewal interval L containing a fixed large time t satisfies $E[L] = E[X^2]/E[X] \geq E[X]$. The interval 'looks longer than average' because you are more likely to land inside a long interval.

- a) Let X take value 1 with probability $1/2$ and value 5 with probability $1/2$. Compute $E[X]$, $E[X^2]$, and $E[L]$. By how much does $E[L]$ exceed $E[X]$ in absolute and relative terms?
 - b) For $X \sim \text{Exp}(\lambda = 1/6)$: compute $E[L] = E[X^2]/E[X]$. Express $E[L]$ as a multiple of $E[X]$. Explain this result using the lack-of-memory property of the exponential.
 - c) For $X \sim \text{Erlang}(2, \lambda = 1/3)$ (sum of two i.i.d. $\text{Exp}(1/3)$ r.v.'s; mean 6, variance 18): compute $E[X^2]$ and $E[L]$. Compare $E[L]$ with the exponential case from (b) and explain the difference in terms of variability.
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Q3 (35 points). Buses arrive at a stop according to a renewal process with inter-arrival times $X \sim \text{Uniform}[4, 16]$ minutes (mean $E[X] = 10$ minutes).

- a) Compute $E[X^2]$ using the formula for $\text{Uniform}[a, b]$: $E[X^2] = (a^2 + ab + b^2)/3$. Find the time-average residual life Y (expected wait from a random arrival time).

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b) If buses arrived deterministically every 10 minutes, the expected wait would be 5 minutes. By how much does the random schedule increase the expected wait? Relate this excess to the variance $\text{Var}(X) = (b-a)^2/12$.

c) A commuter arrives at the stop and observes that the last bus left $s = 2$ minutes ago (age $Z(t) = 2$). Using the conditional formula:

$$P(Y > y \mid Z = s) = P(X > y+s) / P(X > s)$$

compute $P(Y > 3 \mid Z = 2)$ for the Uniform[4,16] distribution.